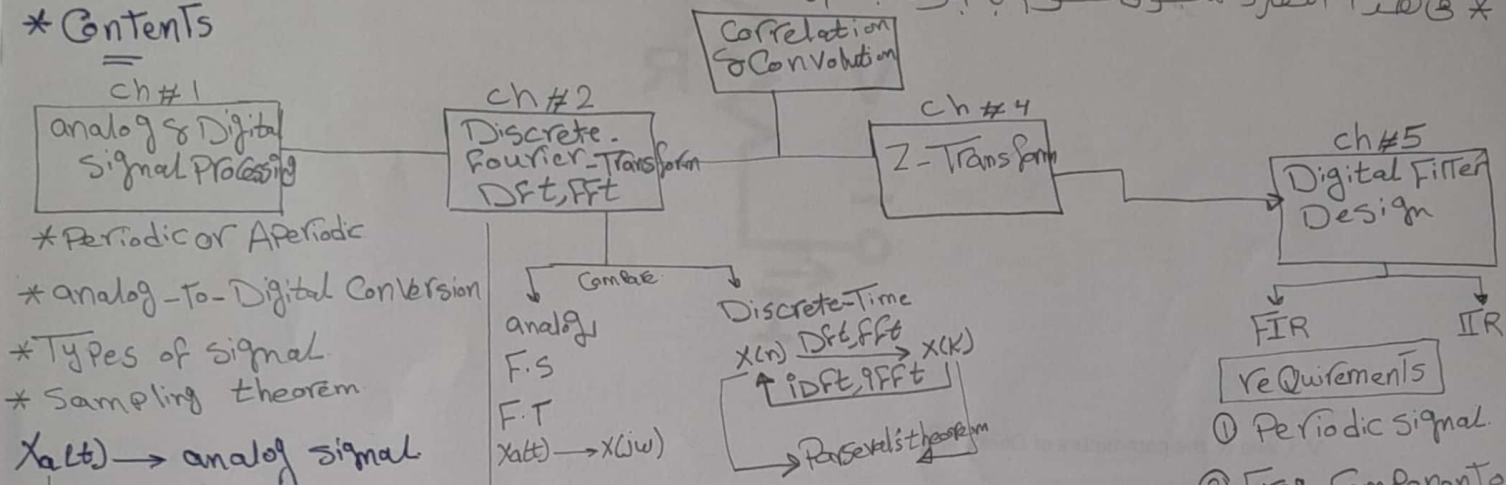
Chapter #1

Digital Signal Processing

المعالجة الرقمية للإشارات

* درسنا في السابق analog signal ← الإشارة الشاخرية وهي الالترانشارا ولكن عند اجراء العمليات عليها تكون غير دقيقة. مثل المقاومة والتأخير في اجراء العمليات. مثل analog Filter
* في هذا المقرر سوف نقوم باجراء بعض العمليات على الإشارة الرقمية مثل الترشيح

* Contents

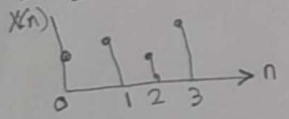


* Periodic or Aperiodic
* analog-To-Digital Conversion
* Types of signal
* Sampling theorem

$X_a(t)$ → analog signal
↳ subscribt
t → Function in Time
Continuous signal

$X_a(t) = A\sin(2\pi Ft + \theta)$
$A e^{xp j(2\pi Ft+\theta)}$

$X(n)$ → Discrete-Time Signal



$X_a(t) \xrightarrow[t \to nTs = \frac{n}{Fs}]{A/D} X(n)$

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analog & Digital signal Processing.

الإشارة والنظم

* Define and give examples.

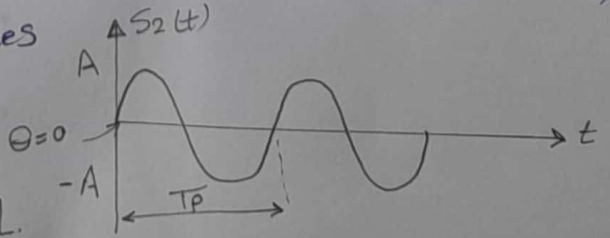
(VI) هي كمية فيزيائية متغيرة زمنيا أو مكانيا ويمكن حمل البيانات.

1 Signal:-
as any Physical Quantity that Varies with Time, Space, or any Independent Variable or Variables

example $S_1(t) = 5t$

$S_2(t) = A\sin(2\pi Ft+\theta)$

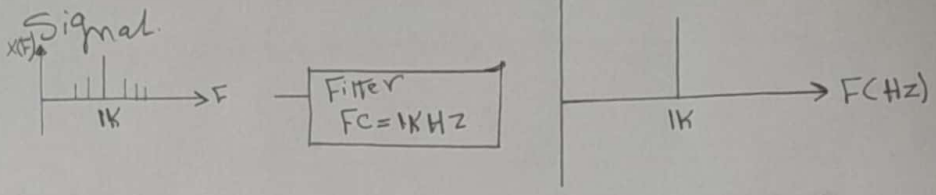
A → amplitude
θ → Phase degree
F → Freq of analog signal.



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2 System: define as Physical Device that perform an operation on the signal.

Example:
Filter

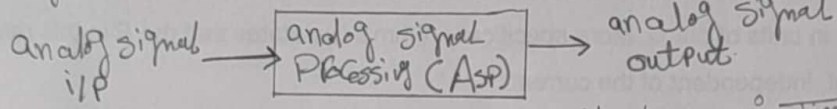


3 Signal Processing:- Set of operation that perform on the signal.

2 Types of Signal Processing

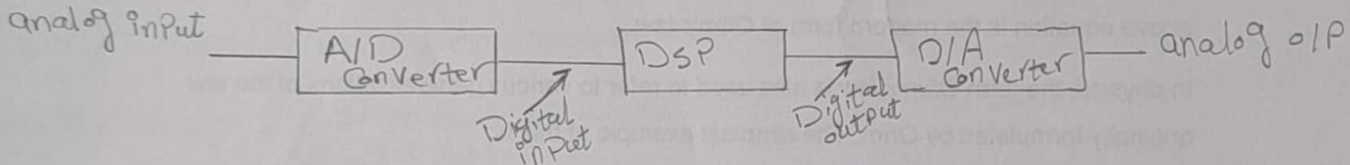
There are Two Types of signal Processing

1 analog signal Processing (ASP)



→ This involves manipulating continuous-signal that vary in time typically using analog ckt

2 Digital signal Processing (DSP)

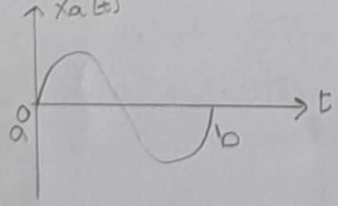


→ In DSP, signal are represented in Digital Form and processed using algorithms implemented on digital Hardware [microprocessors, DSP chips]

3 Types of signal

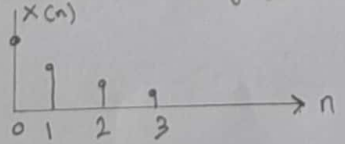
1 Continuous-Time versus Discrete-Time signal

* Continuous-Time:- Continuous-Time signal or analog signal are define for every value of Time and they take on values in the continuous interval [a, b]



$-\infty < t < \infty$
$a < t < b$

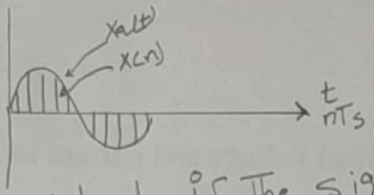
* Disorete-Time: are define only at certain specific values of Time.



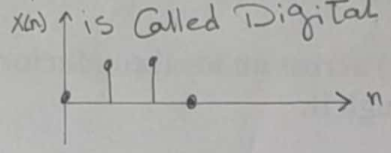
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2 Continuous-Valued versus Discrete-Valued signal

* Continuous-valued
* if the signal takes on values from infinite range.



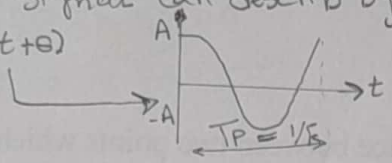
* Discrete-valued:- if the signal takes on values from finite set of values is called Digital signal



3 Deterministic versus Random

* Deterministic:- any signal can describ by mathematical expression, table, figure

$x_a(t) = A\cos(2\pi Ft + \theta)$



* Random:- The signal can't described by mathematical formulas using → Statistical Techniques

∞ Noise

4 multichannel and multidimensional signal

* multichannel:- the signal are generated by multiple sources or multiple sensors

$s(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$ → Voltage, Temp, noise

* multidimensional:- the signal is described by one or more independent variable

$s_1(t) = 3t$

$s_2(x,y) = 2x + 4y$

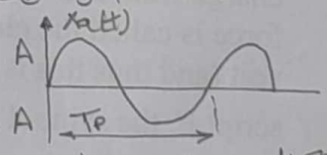
H The main characteristics of the Continuous and Discrete Time signal.

For the Continuous-Time signal.

1 For every fixed value of the Frequency F, $x_a(t)$ is periodic

$x_a(t+T_P) = x_a(t)$

where $T_P = 1/F$ is the Fundamental Period



2 Continuous-Time signal with different Freq are themselves different

3 Increase the Freq "F" results increase the rate of oscillation

$F = 0 \longrightarrow T_P = \infty$

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*For Discrete-Time Signal

$$x(n) = A\cos(2\pi Fn + \theta) \quad -\infty < n < \infty$$

1 Discrete-Time sinusoid is Periodic only if its Frequency F is a rational Number

$$x(n + Np) = x(n)$$

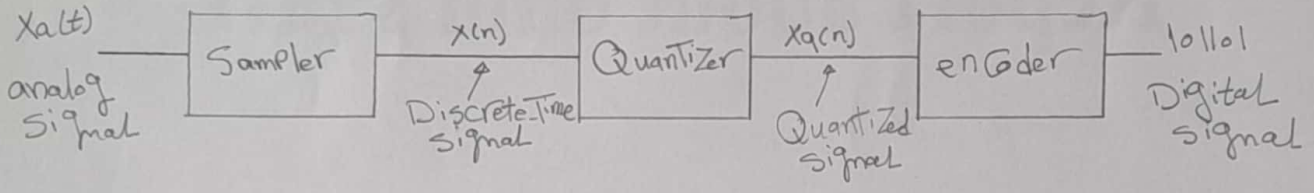
where Np is The Fundamental Period

2 Discrete-Time signal whose FreQuencies are separated by a integer Number of multiple of 2π

3 The Highest rate of oscillation in Discrete-Time is attained when

$$\Omega = \pi \text{ or } F = \frac{1}{2}$$

5 eXPlain The analog-To-Digital Conversion



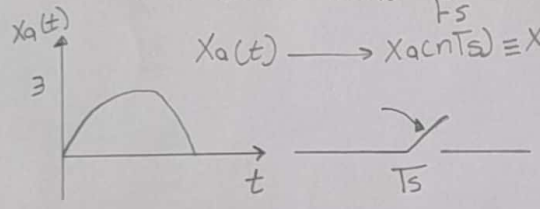
*Sampling:- This is Conversion of a ConTinuous-Time signal into a discrete-Time signal obtained by Taking "samples"

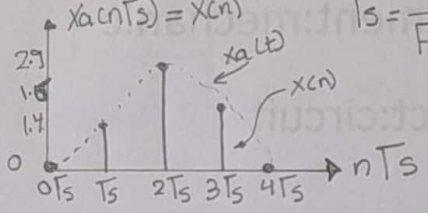
$$t \longrightarrow nTs = \frac{n}{Fs}$$

$$Xa(t) \longrightarrow Xa(nTs) \equiv x(n)$$

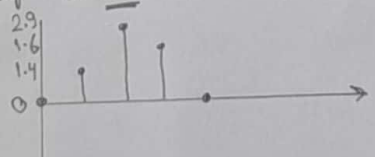
Fs = sampling Freq [S/sec]

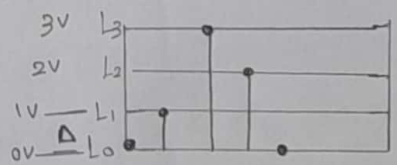
$$Ts = \frac{1}{Fs}$$ Ts = sampling Time





*Quantization: This is Conversion a discrete-Time ConTinuous-Value into discrete-Time, discrete-Value. The Value of each signal sample is represented by Value of selected level





•There are Two method For Quantization Process

$$\Delta = \text{STEP SIZE} = \frac{X_{max} - X_{min}}{L - 1} = \frac{3}{3} = 1V$$

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(*1) Rounding: in This Process the samples Value rounded to The nearest level

* Truncation: is the method used to Quantize sample value by discarding the Fractional Part of the input Values, without Rounding

X(n)	Rounding	error Round	Trunct	error Trun
0V	0	0	0	0
1.4V	1	0.4	1	0.4
1.6V	2	0.4	1	0.6
2.9V	3	0.1	2	0.9

* Quantization error = $e_q(n)$
is The Difference between The un-Quantized sample X(n) and Quantized Signal $X_q(n)$

$$e_q(n) = |X(n) - X_q(n)|$$

To decrease The error

$\Delta \downarrow\downarrow\downarrow$ $L \uparrow\uparrow$ $e_q(n) \downarrow$

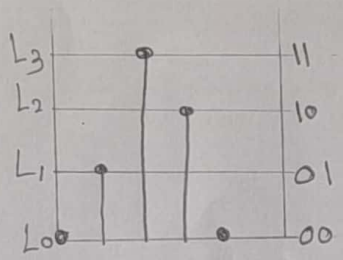
$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}$$

[3] enCoding:- in This Process, each Discrete Value $X_q(n)$ is represented by b-bit binary seQuanCe

$$L = 2^b$$

where L ⇒ Number of level
b ⇒ Number of bit

For L = 4 level $4 = 2^b$
b = 2 bit



00 ▽ 01 ▽ 11 ▽ 10 ▽ 00 Digital signal

* True or False and Correct

① in Quantization Process, when The Quantization step (Δ) increase, the accuracy of the Quantizer increase (X) → the accuracy of Quantizer decrease

② Ny Quist Freq (FN), is the minimum sampling Freq (✓)

③ To reConstruct The Continuous Time signal, x(t) From its discrete Time sequence, x(n), the sampling rate Fs, must be less than the Freq of analog signal (X) → must be greater than or equal Twice of analog Freq

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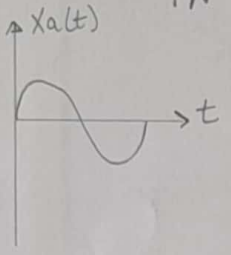
6 explain The sampling theorem.

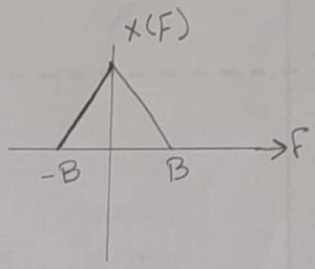
For The analog signal $x_a(t)$ with maximum Freq (F_{max}) Can Sampled this Signal with sample rate (F_s) must be greater Than or eQual Twice The max Freq of analog signal To reConstructed The original Signal From it's Sample.

$$F_s \geq 2F_{max} \quad (\text{without aliasing})$$

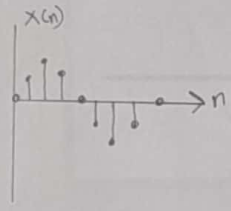
*Ny Quist rate (F_N):- it is The minimum samde rate To samples the analog signal without aliasing

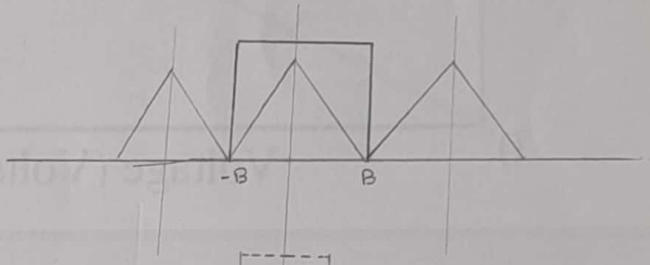
$$F_N = F_s = 2F_{max}$$



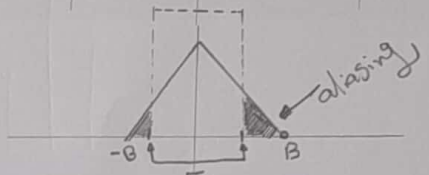


aFTer Sample

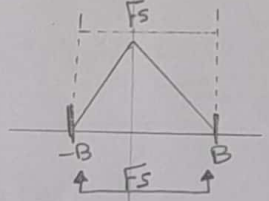




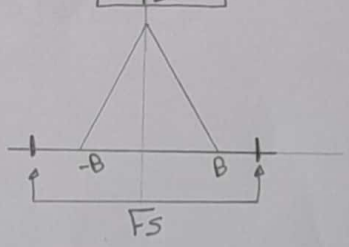
* For $F_s < 2F_{max}$



* $F_s = 2F_{max}$ = Ny Quist rate.



* $F_s > 2F_{max}$



You must select $F_s \geq 2F_{max}$ To reConstrucoted The original signal

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⑦ Drive The Relation beTween analog Freq (f) and Discrete (F) and the sampling Freq (Fs)

The General Form of analog signal

$$X_a(t) = A\sin(2\pi f t + \theta) \longrightarrow ①$$

at sampling Process

$$t \longrightarrow nT_s = \frac{n}{F_s}$$

$$X_a(nT_s) = X(n) = A\sin\left(2\pi f \cdot \frac{n}{F_s} + \theta\right) \longrightarrow ②$$

The General Form of Discrete-Time

$$X(n) = A\sin(2\pi F n + \theta) \longrightarrow ③$$

by Compare ② & ③

$$F = \frac{f}{F_s} \quad (F_s \& f \text{ is linear Relation})$$

where

A → Amplitude [v]

θ → Phase shift [rad]

f → Freq of analog → [cycle/sec]

Fs → Sampling Freq [sample/sec]

F → Freq of Dis [cycle/sample]

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⑧ Prove That the analog signal is Periodic Signal

The General Form

$$X_a(t) = A\sin(2\pi f t + \theta)$$

For The Periodicity

$$X_a(t+T_P) \stackrel{?}{=} X_a(t)$$

$$X_a(t+T_P) = A\sin(2\pi f(t+T_P) + \theta)$$

$$= A\sin(2\pi f t + \underbrace{2\pi f \cdot T_P}_{2\pi} + \theta)$$

$$2\pi f \cdot T_P = 2\pi$$

$$\boxed{T_P = \frac{1}{f}}$$

$$= A\sin(2\pi f t + \theta) = X_a(t)$$ is Periodic Signal with Fundamential Period $T_P = \ldots$ sec.

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⑨ Prove that the Discrete-Time Signal is Periodic.

$$X(n) = A\sin(2\pi F n + \theta)$$

For the Periodicity

$$X(n+N_P) = X(n)$$

$$X(n+N_P) = A\sin(2\pi F(n+N_P) + \theta)$$

$$= A\sin(2\pi F n + \underbrace{2\pi F N_P}_{2\pi K} + \theta)$$

$$2\pi F N_P = 2\pi K = \text{integer Number}$$

$$F = \frac{K}{N_P} = \frac{\text{cycle}}{\text{samples}} \longrightarrow K, N_P \text{ must be integer Number}$$

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⑩ Find The Relations between The Freq of Discrete, analog signal

Continuous-Time		Discrete-Time
$\omega = 2\pi F$ radians/sec Hz		$\Omega = 2\pi F$ rad/sample · cycle/samples
	$\Omega = \omega T_s$, $F = \frac{f}{F_s}$	$-\pi \le \Omega \le \pi$ $\frac{-1}{2} \le F \le \frac{1}{2}$
$-\infty < \omega < \infty$ $-\infty < f < \infty$	$\omega = \Omega/T_s$, $f = F \cdot F_s$	$\frac{-\pi}{T} \le \omega \le \frac{\pi}{T}$ $\frac{-F_s}{2} \le f \le \frac{F_s}{2}$

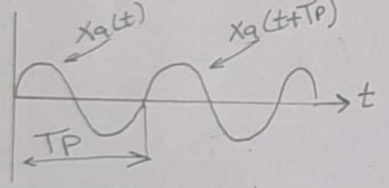
- To check The Periodicity

① For analog signal.

$$x_a(t) = A\sin(2\pi F t + \theta)$$
$$= A\cos(\quad - + -)$$
$$= Ae^{xpj}(\quad - + -)$$

The Condition For Periodic

$$x_a(t+T_P) = x_a(t)$$



- The Fundamental Period T_P

② For Discrete-Signal

$$x(n) = A\cos(2\pi F n + \theta)$$

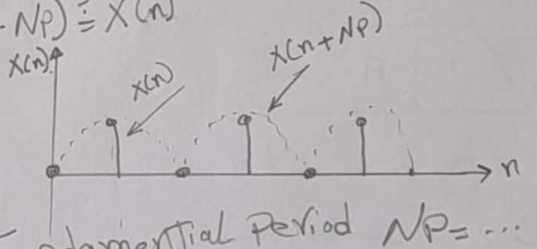
$$F = \frac{K}{N_P} = \frac{\text{cycle}}{\text{sample}}$$

→ Cancel the Common Factor $F = \frac{2}{6} = \frac{1}{3} = \frac{K}{N_P}$

→ must K, N_P integer Number

The Condition

$$x(n+N_P) \stackrel{?}{=} x(n)$$



- The Fundamental Period N_P = ... Sample/cycle

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